

UNIT - V

INSTRUCTIONAL MEDIA

5.1. Print resources:

Newspapers, books, journals, magazines, encyclopedias, etc. are called print resources. They give variety of information and contain factual information.

5.1.1. Newspapers

Newspaper is a rich source of information. It communicates authentic and first-hand information. Any information was immediately reported by the newspapers. Photographs, recent scientific information, new strategies are all published by the newspapers. Newspapers consist of the following news.

1. General news
2. Political news
3. Education news
4. Area news
5. World news
6. Sports news
7. Weather news
8. Job news
9. Advertisement news
10. Breaking news

Contents of the newspaper items relates to the school curriculum. So the teacher can use the news in classroom and may use a bulletin board to display the newspaper cuttings on a particular topic. The display news could be changed every day. Students can be given the task

of writing daily importance news items on the chalkboards in the corridor.

Student can prepare files of newspaper cuttings on different topics or issues. These cuttings can then be used for conducting group discussions, giving group assignments or even for individual study. These files come handy when you require information on a particular aspect of a topic. The files prepared by the teacher and students can be maintained in the school library.

Benefits

- Newspapers are an adult medium and students all ability levels are inherently proud to be seen reading them.
- Newspapers deal with contemporary issues which are happening here and now and hence provide motivation for reading and discussion.
- Newspapers make learning 'fun'.
- Newspapers are extremely flexible and adaptable to all curriculum areas and grade levels.
- Newspapers bridge the gap between the classroom and the "real" world.
- Newspapers build good reading habits that will last a lifetime.
- Newspapers can be cut, marked, clipped, pasted, filed and recycled.
- Newspapers give everyone something to read – news, sports, weather, editorial and comics.
- In the long run, newspapers are cost effective and provide education.
- Newspapers contain practical vocabulary and are the best models of clear, concise writing.

5.1.2. Journals

A journal is a magazine that deals with a specialized subject. Many magazines have Journal as part of their name. A journal is also a kind of diary in which you keep a record of events or progress. Journal has several related meanings.

- A daily record of events or business; a private journal is usually referred to as a diary.
- A newspaper or other periodical, in the literal sense of one published each day.
- Many publications issued at stated intervals, such as academic journals (including scientific journals), or the record of the transactions of a society, are often called journals. In academic use, a journal refers to a serious, scholarly publication that is peer-reviewed. A non-scholarly magazine written for an educated audience about an industry or an area of professional activity is usually called a trade magazine.

Public journal

A public journal is a record of day-by-day events in a parliament or congress. It is also called minutes or records. Journals also found in Thesaurus, Medical, Legal, Financial, Acronyms, Encyclopaedia, Wikipedia.

Journal of Biological Science is aimed to publish research articles on theoretical foundations of information and computation and of practical techniques for their implementation and application in Biological systems. JCS (Journal for Biological Science) updated twelve times a year and is a peer reviewed journal covers the latest and most compelling research of the time

E-journals

The journals which are web-based and available online are known as electronic journals or e-journals. It is based on the web based technology. It is the one of the significant applications of Biological in the field of education.

There are varieties of journals in internet. We can easily access it. In internet, there are two sources of journals. One is title. Another one is full text. If the title is available online, it may provide table of contents, tables of contents with abstracts or the full text of the journal. The full text is provided either HTML(Hypertext Markup Language) format or PDF format(Portable Document Format).

Some e-journals

- Anthropological Linguistics
- Sintagma
- Translation journal
- Pedagogical of quarterly of cognitive linguistics
- Journal of S.I.T.U. Council of Educational Research

Advantages

- They are inexpensive.
- Journals can be accessed immediately and very easily.
- We can easily download.
- Free journals are available.
- We can access the journal from everywhere.

5.1.3. Magazines

A magazine is something that lies between a newspaper and a book. Unlike the book it is published periodically in a series. But it differs from a newspaper. The reporting of day-to-day happenings is not its main concern. To inform, instruct and entertain through comments on chosen news and features on subjects of interest is what it is engaged in.

The comments are not cursory and hastily put down as in a newspaper as there is relatively longer time lag between the story happening and publishing. The time lag gives ample time for putting together well thought of and synthesized reactions. Newspapers are read on the day and thrown away the next day. Magazines are read and reread and passed around to others or stored away for future reference. They travel far and reach a wider audience. There are many types of magazines. They are as follows.

- New Magazines
- Sports Magazines
- Cultural Magazines
- Intellectual Magazines
- Magazines for children
- Magazines for men
- Magazines for women
- Business Magazines
- Educational Magazines

Educational Magazines

- It provides a variety of reading materials which exposes children to literary genres including poetry, photo stories, short stories, fiction, non-fiction and current event articles.

- It supplements classroom instruction by providing additional material in different subjects.
- It includes writing activities that reinforce reading skills and improve grammar, exercises that develop spelling, vocabulary and word analysis skills, puzzles and riddles that develop critical thinking and reasoning power and activities for numerical data processing and chart-reading skills.
- It leads to professional development of teachers in classroom management techniques, subject knowledge and instructional practices and strategies.
- It provides opportunities for independent student reading practice and at home reading assignments.
- It provides a means of proper utilization of leisure time for students and act as source of entertainment.
- It enriches the learner with news about the outside world and heritage and culture.
- It can foster moral and spiritual values and develop national and international understanding.

5.1.4. Biological Science encyclopedias

Encyclopedia is a type of reference work or compendium holding a comprehensive summary of information from either all branches of knowledge or a particular branch of knowledge. Encyclopedias are divided into articles or entries, which are usually accessed alphabetically by article name. Encyclopedia entries are longer and more detailed than those in most dictionaries. Generally speaking, unlike dictionary entries, which focus on linguistic information about words, encyclopedia articles focus on factual information concerning the subject for which the article is named.

Encyclopedias have existed for around 2,000 years. The modern encyclopedia evolved out of dictionaries around the 17th century. Historically, some encyclopedias were contained in one volume, whereas others, such as the *Encyclopedia Britannica*, the *Encyclo+pedia Italiana* (62 volumes, 56,000 pages) or the world's largest, *Enciclopedia universal ilustrada europeo-americana* (118 volumes, 105,000 pages), became huge multi-volume works. Some modern encyclopedias, such as Wikipedia, are electronic and often freely available.

The *Encyclopedia of Science* is the definitive reference in science and technology. It is a reference book for Physics, Chemistry, Botany and Zoology. Beginning with the Greeks and Hipparchus, continuing through Ptolemy, Copernicus and Galileo and today our work on the International space station man continue to learn more. This influential work provides an historical timeline highlighting the key breakthroughs in science and technology, as well as clear and concise explanations of the latest technology and its practical applications. Its unique blend of historical perspective, current knowledge and predicted future trends has earned it its richly deserved reputation as an unrivalled reference classic.

What sets the *Encyclopedia* apart from other reference sources is the comprehensiveness of each of its entries. Encompassing far more than mere definitions, each article elaborate on a topic giving a remarkable breadth and depth of coverage. The visual impact of the volume is enhanced with a 16 page colour insert spotlighting advanced Biological applications and Biological-generated graphics technology. In addition, the text is enlivened with figures, tables, diagrams, illustrations and photographs.

5.2. Audio resources

Radio resources like Radio, Tape recorder, CD etc. enhance teachers ability to present the lesson in simple effective and easy to understand for students.

5.2.1. Radio talk

Radio talk is a one-way communication media. Radio talks are a very cost effective. It gives only auditory experience. Radio is the transmission and reception of signals helps by means of electronic waves. Radio listening helps the student to add many information in student's knowledge bank. It exerts an influence on student's attitudes, appreciations, social thinking.

Radio is an educational aid. It provides valuable auditory experience simultaneously to a large number of learners. Many specially arranged talks and discussions and information on biology are broadcasted by the radio station in their scheduled institutional or general mass educational programmes. These programmes are very useful for students. Every educational institution should know when they will broadcast these programmes. That time they are ready to hear the programmes.

Talk radio is a radio format containing discussion about topical issues. Most shows are regularly hosted by a single individual, and often feature interviews with a number of different guests. Talk radio typically includes an element of listener participation, usually by broadcasting live conversations between the host and listeners who "call in" (usually via telephone) to the show. Listener contributions are usually screened by a show's producer(s) in order to maximize audience interest and, in the case of commercial talk radio, attract advertisers.

1. Microphone
2. Amplifier
3. Reproducer

It involves two main processes.

1. Recording

In recording, the educational message is first fed into the tape recorder through microphone and other inputs. The voice produces mechanical vibrations which are changed into electrical vibrations. The amplifier intensifies these vibrations which in turn activate a magnet. It results in the establishment of varied sound pattern on the iron oxide coated tape.

2. Reproducing of the recorded sound

The playing of the instrument results into the reproduction of the recording sound. Here the sound patterns contained on the tape are subjected to electrical vibrations which are amplified by the amplifier and changes into an original like voice by the speaker.

Advantages

- It is quite helpful in the learning of some special subject like music, dramatics and language etc.
- It helps to avoid poor speech habits and to correct actual speech defects.
- It helps to develop the conversation skill, expression power and techniques of effective dramatization.
- It is used to evaluate the teaching and learning programme.

- It is used to record the various seminars, workshops and conference.
- It helps a teacher for assessment of own behaviour and teaching task.

5.2.3. DVDs /CDs

DVD ("digital versatile disc" or "digital video disc") is a digital optical disc storage format. It was invented and co-developed by Philips, Sony, Toshiba, and Panasonic in 1995. The medium can store any kind of digital data and is widely used for software and other Biological files as well as video programs watched using DVD players. DVDs offer higher storage capacity than compact discs while having the same dimensions.

Pre-recorded DVDs are mass-produced using molding machines that Biologically stamp data onto the DVD. Such discs are a form of DVD-ROMs, because data can only be read and not written or erased. Blank recordable DVD discs (DVD-R and DVD+R) can be recorded once using a DVD recorder and then function as a DVD-ROM. Rewritable DVDs (DVD-RW, DVD+RW, and DVD-RAM) can be recorded and erased many times.

DVDs are used in DVD-Video consumer digital video format and in DVD-Audio consumer digital audio format as well as for authoring DVD discs written in a special AVCHD format to hold high definition material (often in conjunction with AVCHD format camcorders). DVDs containing other types of information may be referred to as DVD data discs.

Digital Versatile Disc/ Compact Discs are available to record visual programmes with sound on lasers. Any information can be easily read from any point on the disc. It is used to record television programmes or self-produced programmes by the video camera and exhibited to the students after some time. We can record important

educational telecasts outside the school time and it can be played in the classroom teaching. They are used to record our own programme and play it to the class for the future improvement. We can record a lecture or talk given by the experts and played back after sometime.

5.3. Visual resources

Visual resources like pictures, charts, posters, photographs, flashcards, models etc. enhance teachers ability to present the lesson in simple, effective and easy to understand for students. This resources make learning more permanent since student use more than one sense.

5.3.1. Pictures

Pictures occupy an important place in visual aids. Magazine pictures are easily and cheaply available from old magazines, calendars, newspapers, pamphlets and so on. They are more imaginative than the commercially produced wall charts and flash cards. The following points should be remembered while selecting pictures. They are:

- Cost
- Clarity
- Suitability (big or small)
- Classifiable from the subject or syllabus point of view
- Limited or a variety of uses.
- A ready-made magazine picture album may be built up, in order that the teacher may not search for them every time he needs them.

Uses of Pictures:

- Motivation can be given using pictures.
- To present new vocabulary, structures and situation incidentally.

- For drills and pair work activities.
- For listening and reading and picture composition.
- Pictures of Scientists and Writers can make the student familiar with them.
- A series of pictures can be used for a series of connected actions to teach an item.

5.3.2. Flash cards

Flash Cards are wonderful teaching tools. They offer fun, interaction learning for all levels of education for all subjects. Flash Cards created by teacher or purchased from stores it can help with memory skills, mathematical skills, interaction, speaking and listening skills of all ages. They give students instant feedback and correction.

A flash Card is a rectangle shaped piece of card, Biological Science formula or something is written or some picture is drawn and flashed for a short time. Waste chart, waste greeting and invitation cards may be cut to neat sizes usually 3"x10" size. Biological Science formula, few steps may written under them.

We can use flash Cards in two different ways. In first type questions will be in one side and its answer will on the backside of the same card. In second type questions will be on separate cards and answers are also written on separate cards. After completing a lesson, teacher can prepare 15 to 20 flash cards and review the lesson and utilize it as teacher-student, student-student form to review the content.

You can make flash cards using with microsoft words' templates. Flash card software is becoming increasingly popular due to new technology.

Flash Card Tips

- As the name suggests, what you need to do is simply flash cards in front of students. One important condition is to make sure they are watching. Show the card and the students have to tell what is on it.

Uses of the flash Cards

- Some questions or steps are written on the flash cards. Students read them and answer the questions.
- can be conducted with the help of flash Cards.
- They brings more motivation
- It enables the students to recognize computer science concepts, formule etc.
- Flash cards engage active recall.
- Flash cards help to assess correctness of the answer. This act is known as Self - reflection.
- They are inexpensive.
- They are portable.
- They are efficient.
- They make learning easier.
- They are versatile.
- They offer various study methods

Disadvantage of flash cards

- Some lexical fields cannot be illustrated.
- They can become boring when using repetively.

5.3.3. Charts

Charts are one of the most commonly used teaching aids in schools. The chart is a systematic arrangement of key facts or ideas in a logical sequence or representing ideas and facts in a pictorial or graphic or graphic form. The charts can be used for a variety of purpose such as

- motivating the students
- deriving formula and to explain
- Showing important key points in a lesson
- To understanding the abstract mathematical concepts
- Summarizing a lesson
- Showing steps in proving theorems
- Depicting interrelationship among mathematical concepts.
- It helps development of scientific attitude.
- It gives direct experiences

The use of charts save teachers time and labour. The charts help the teacher to capture the attention of the students.

Normal size of a chart is 51 cm x 64 cm. It is a two dimensional visual aid. Before using charts teachers should ask themselves these questions.

- Is the content content relevant to my topic? Do the charts meet my lesson objectives?
- Is the information accurate? Are the facts clearly presented?
- Can my students understand these visual symbols?

- Are Charts attractive and motivate the pupils in their learning.

Only when the answer to these questions are in the affirmative, use charts for instruction.

Types of Charts

- Picture chart
- Time chart
- Graphic chart
- Flow chart
- Tree chart
- Bar chart
- Pie chart
- Flip charts - It is a collection of charts which are bound together at the top.

5.3.4. Posters

A poster is a display material especially prepared to convey a message or an idea and to create a strong lasting impression. The effectiveness of poster is based on the design, effects and cohesion of ideas. The purpose of a poster is to

- Motivate class interest.
- To present a contemporary relevant issue.
- To convey message accurately and carefully.
- To arrest attention by using attractive features.

Class room posters are simple, yet effectual approach for learning that will help all levels of education.

There are five points to make a poster

- The theme idea - Clearly stated
- Way of presentation - in a simple and bold in letters
- Wording - Should be simple and pleasing
- Design of the posters - Should be attractive
- Colour management - Use attractive colours

Uses of Posters

1. They add visual representation.
2. They allow for peer to peer interaction.
3. Students can learn from each other's posters.
4. They develop a wide range of key transferable skills.
5. Posters help reinforce important concepts, skills
6. Images, Photographs and diagrams are helpful learning tools for visual students.
7. Words linked to poster help students grasp and remember new concepts.
8. About one third of all students in an average classroom learn through visualisation.
9. It is an independent learning aid.
10. Posters provide great teaching material that you can refer to during teaching.

5.3.5. Photographs

There are various methods for making the curriculum more interesting, enjoyable and student centred. Photographs make students are meaningful and interesting. A photograph is worth a thousand words through which a complex idea can be conveyed with just a single still image. Photographs make it possible to absorb large amount of data

quickly. Using photographs for explaining complex phenomena is one of the teaching aid of modern education system all over the world. As the world is changing day by day so as the methods of instruction as the modern curriculum. Use of photographs is important for students because they are more likely to believe findings during learning. Using photographs may be helpful in arousing interest in a new topic. Photographs can be used promote positive learning experiences for students when used in ways that are consistent with the cognitive theory of visual learning. Therefore photographs have an added advantages over text.

Photographs can be lively engaging and effective tools for learning the class room. We live in a way visual world where images are used in numerous situations to carry messages and communicate ideas. Students are much more comfortable with visual images then with verbal ones, a combination of visual and verbal can enhance student interest and involvement.

Objectives of using Photographs in teaching learning process

1. Enhancement in perception level of students.
2. Easy interpretation of context.
3. Smooth understanding by learners.

Activities for using Photographs

In exploring students understanding the teacher could use the following activities:

- * Encourage students to develop their own collections of Photographs in a folder relative to Biological Science.

Show a photograph of a Biological Science content and ask the following questions:

- * Describe what you see in the photograph
- * What is the main subject matter of this photograph?
- * How would you classify this photograph?
- * What is the overall meaning of the photograph?

Uses of Photographs

1. It helps in motivating a new topic.
2. Photographs of eminent mathematicians can make the student familiar with them.
3. To know how Biological Science concepts are used in real life situations.
4. We can interact with students
5. It helps to develop student teacher relationship.

5.3.6. Models

Models are the replicas or copies of the real objects. Models are recognizable 3 dimensional representations of real things. It also attempt to provide interior views of objects which are normally covered or otherwise invisible. They are usually of three types

i. Scale models

Scale Models are made with same size and shape of the object.

ii. Cross-sectional models

Sectional models explain clearly the structure or functions of the original.

iii. Working models

It is used where the specific function of the original is duplicated and could be explained easily.

Functions of Models

- It simplifies reality.
- It concertises abstract concepts.
- It enables us to reduce or enlarge objects to an observable size.
- Models are three dimensional objects. So it is better than charts to explain subject matter clearly.
- It explains the various processes of objects and machines.

Criticisms of the use of models

- Can be too linear
- Not flexible enough, too rigid.
- Very costly
- Rarely used in the real world
- Only applies to me particular situation.

5.4. ICT resources:

ICT can make a significant contribution to teaching and learning more effective in all subject areas. It encourage and support collaboration between teachers and students. Teachers need specific professional development opportunities in order to increase their ability to use ICT equipments.

5.4.1. Radio

Radio is an effective audio aid device. It is capable of providing valuable assistance to the teacher in the classroom by presenting worthwhile information and learning experience simultaneously to a large number of students.

Radio is the most significant medium for education. As a supplement to classroom teaching its possibilities are almost unlimited. Its teaching possibilities are not confined to the five or six hours of the school day. It is available from early morning till long after midnight. By utilizing the rich educational and cultural offerings of the radio, children and adults in communities, however remote, have access to the best of the world's stores of knowledge and art. Someday its use as an educational instrument will be as commonplace as textbooks and blackboards.

- **R.G. Reynolds**

Radio is the transmission and reception of signals by means of electromagnetic waves. Radio listening helps the students add knowledge bank. It exerts an influence on student's attitudes, appreciations, social thinking.

In many states of India All India Radio (AIR) broadcast school programmes. The school radio generally broadcasts programmes based on primary and secondary school curriculum. They are broadcast in the state language. A copy of the broadcast schedule for a term/semester and the syllabus (i.e. content covered) are sent to schools so that schools can plan their time-table accordingly.

The school broadcast can be used and are in fact being used in two ways. One is directly at the time of broadcast and second is by recording the programme on audio cassette and using it whenever convenient. The audio cassette technology gives more control to the users – the teacher and the student. A catalogue of such taped programmes can be made for easy retrieval and use.

Audio cassettes/radio programmes are generally prepared on topics which are more suitable to verbal communication. Sound, music and special audio effects can be used in audio programmes as to make them more effective. These techniques help create visual images through sound.

Uses of Radio

1. It is inexpensive as it can reach a wide audience.
2. It is an instrument that can bring the information live and hence is up-to-date.
3. It reduces the barriers of time and space.
4. It improves the speaking skills.
5. It stimulates a spirit of co-operation and responsibility.

5.4.2. Television

Television is the most powerful medium of communication. It has revolutionized the method of teaching and learning. It is a convenient and economical medium of reaching a large cross-section of population. It combines the best of radio and motion pictures. It helps overcome barriers to learning. There are various programmes which provide up-to-date news around the world like Discovery Channel, Animal planet, BBC, CNN, National Geographic, ESPN, Ten Sports, Pudiya thalamurai, Doordarshan and so on.

Now-a-days, it has become the most recent audio-visual aid for classroom instruction. This is a multi-sensory audio visual aid, which can be effectively used for language teaching. The Directorate of School Education, Government of Tamilnadu, Chennai and the Doordarshan Kendra, Chennai has been telecasting programmes for teachers and students.

There is a recent development in Educational television programme. They are:

- Satellite based communication
- Telephone clubbed with television
- Video tapes and
- Multimedia packages

Educational Television

The television is called as the "queen of audiovisual aids". It is a one of the mass communication. It appeals to both the eye and the ear. There are three types of educational television programmes.

1. Direct teaching enrichment programmes
2. Supplementary enrichment programmes
3. Demonstration type programmes

There is a recent development in Educational television programme. They are:

- Satellite based communication
- Telephone clubbed with television
- Video tapes and
- Multimedia packages

Uses

- A large number of students can be given information at a time.
- It can help supplement class room oral teaching.
- It develops in them positive and self-confidence towards the foreign language by listening and seeing the English channels.
- Television can be used for telling stories, teaching of structures and sentences, their forms and patterns.
- It also inculcates and develops a sense of critical appreciation among the young learners by showing correct and modulated poetry reading.

- Television programme can highlight the function of the speech organs in producing right speech sounds.
- Listening, speaking and understanding abilities of the learners can be improved.
- The students who are not able to attend the class can watch and listen from their homes itself.
- It is time saving. Most of the syllabus may be covered in less time.
- Doordarshan programmes like UGC, Kanbom karpom are very helpful for students who are appearing for their public exams as the lessons and the question models are explained in detail.

Disadvantages

It is a one-way communication only.

Slow learners cannot cope up with it.

- Again there is a dearth for teaching who can explain the students before the programme commences.
- Poor accessibility and it is difficult to integrate the students.
- Sometimes visual becomes a distraction in the learning process.

5.4.3. Internet

Internet is the most user-friendly graphical system that offers huge amount of information to the users. Internet reveals huge sets of pages containing data, information, images, symbols, words, sounds, video clips etc., collected from all over the world. Internet is a network of networks. It is a group of two or more networks that are

- Interconnected Biologically
- Capable of communicating and sharing data with each other and
- Able to act together as a single network.

We can exchange text, data files, messages and programs with any other user.

Significance of internet

1. Procuring information on the internet

The information about people, cultures, societies, languages, nations, products, organizations, inventions, discoveries, games, events, research data and findings, health and nutrition, developmental activities, media, politics, day-to-day news etc. is available on the internet.

2. Providing information on the internet

Individuals and institutions would like to re-present their ideas, praises, activities, programs, new-proposals, mission and vision statements, services offered, interests etc. over the internet. They can set up their home pages in a decorative and attractive way to extend their market and services. For this purpose, one can take up publishing, extension, teaching etc., on the net.

3. Compiling information on the internet

Internet can be used for obtaining specialized information from the web. One can generate opinions of people, experts, specialists, politicians, religious leaders, etc. by administering various instruments through e-mail or involving in group discussion, list servers, chatting groups, job groups, medical experts groups, etc.

Internet in Education

The internet has very wide applications in the field of education. It provides a number of learning experiences and educational resources to students, teachers, administrators, policy makers and all those who work for the cause of education. Some of the important ways in which the internet helps children are

- It helps develop and improve reading, writing, research and language skills.
- It provides support for kids with special needs and their parents.
- It is an exciting outlet for artistic expression.
- It provides information about every topic imaginable.
- It is a great way to hang out with your friends and be at home at the same time.
- It provides personal contact with new people and cultures.

The open learning methodology means the learners need not to go to the educational institutions. The internet provides a huge amount of information to the students across the world. So the internet can be used for delivering instructional inputs or providing information to the open school learners.

Internet will provide information to the schools and students in the following areas.

- Access to electronic information. Archives and databases.
- Direct support to teachers and learners to enhance educational access, quality.

- Career counselling
- Job requirements and opportunities available.
- Courses available through distance education and open learning mode.
- Provision of internet services.
- Provision of directory services.
- Offering courses through internet.

5.4.4. Multimedia

Multimedia is used in different types of fields. In general Multimedia method is used in Entertainment, Education, Research and commerce departments. Multimedia Educational Software has done large effective in Education. Many powerful Electronic Education Software collections are created by using the growing multimedia facilities. These are used to teach the students in globally. These are presented with the self-learning facility by the Internet and Compact Disc. Multimedia approach is one of the innovative methods that are aimed at improving the teaching – learning process. Some of the Multimedia Usages in Teacher Education are as follows

Definition

Multimedia is a hot topic in current education because it represents the latest technology and introduces into the classroom whole new ways of thinking about curriculum interactions with students and even the nature of learning itself. He elaborate that the meaning of multimedia has changed from meaning nothing to everything. Multimedia can mean any kind of file or document either a text or spreadsheets that have audio/video effects or “an interactive information cafe”. Whatever it is not, it certainly is the most promising technology in education.

- **WeidongXhang (2003)**

Usage of Multimedia in Teacher Education

1. Power Point Presentation

We can use the PowerPoint presentation to teach the lessons in effectively. It consists of many elements. Some of them

1. Text
2. Images
3. Pictures
4. Links
5. Animation and etc.,

2. CAI Packages

It is one of the individualized instructions. Student can interact with the Biological to learn the lessons. The teacher's role is only giving the instructions. These packages are instructional techniques using the Biological which follows for individual, individually paced and individualized instruction. Since the Biological's behaviour is dependent upon the responses of the student. In CAI package, the information is presented in a structured form. It can provide a method of instructions designed for self-directive study. It helps in improving skills or achieving objectives at all difficulty levels. It provides immediate feedback. It saves the unauthentic labour of teachers as well as students.

Devices used in CAI:

1. Biological
2. Storage devices like; floppy, C.D.,
3. Film Projector
4. Multimedia Projector, LCD,

5. Input devices like; light pen, voice synthesizer, microphone, web camera etc.
6. Speakers

3. Mobile Learning

Mobile learning is one of the Multimedia Learning. It motivates to learn latest techniques or knowledge. It provides mobility of learning setting; not restricted to classroom learning. It consists of interactivity of the learning process and we can interact with others. It gives situational aspects of instructional activities. It integrates the instructional content all in one.

4. Smart Classroom

Smart classroom is a new type of classroom which replaced the formal and traditional classroom environments. It focuses on student's intellectual development not mere learning.

Requiring Equipments

A smart class is a classroom that has an instructor station equipped with Biological and audio-visual components. Required equipments for smart classroom are as follows.

- ❖ Personal Biological
- ❖ Overhead projector
- ❖ Wireless Internet Access
- ❖ DVD player
- ❖ Smart board
- ❖ Speakers

These components are connected wirelessly, via USB or serial cables.

Contents are illustrated through animations, PowerPoint presentations and videos. Additional notes are added with the help of relevant internet sites. Students are shown keen interest towards class and respond in a well manner. Experts in various fields can bring into classroom through video-conference. The interactive nature of the smart notebook software, teachers can record each step of a lesson activity for student to review at a later time.

5. E-Learning

E – Learning includes numerous types of media that deliver text, audio, images, animation and streaming video and includes technology applications and processes such as audio or video tape, satellite TV, CD-ROM and Biological based learning as well as local intranet/extranet and web based learning. It is use of electronic media and information and communication technologies (ICT) in education.

Procedure for adopting Multimedia Approach

Stage 1: Teacher initiates teaching learning activities.

Stage 2: Teacher demonstrates a specific and specialized unit.

Stage 3: Preparation and ground work for students to embark on independent learning.

Stage 4: Give the student's active participation.

State 5: Integration of theory and practical.

Role of Teacher in Multimedia Approach

- The teacher has to adopt a number of methods and techniques he cannot be satisfied with the lecture or the chalk talk method.
- The teacher has to be aware of the different available media and their availability.
- The teacher should be Biologically component to use and demonstrate the use of different media.
- The teacher should provide experience such that the students can link practice and theory and integrate them.

Advantages of Multimedia

1. Multimedia can be helpful to organize information in meaningful ways.
2. It offers the different learning styles.
3. It allows for self-pacing and discovery.
4. It helps to develop the high order thinking skills.
5. It provides the 'anywhere', 'anytime' learning.
6. It helps in developing group and interpersonal skills.
7. It encourages collaborative learning and enhances the interactions.
8. It helps the students to learn the content in a given discipline.
9. It helps students to learn to think effectively, practice problem solving and decision making.
10. It can bridge language barriers for audio is not only means of communications.
11. It enables students to represent information using several different media.
12. It provides better communication between students via e-mail, chat session etc.,

5.4.5. Interactive whiteboard

Interactive board is an instructional tool that allows Biological images to be displayed on to a white board using digital projector. The instructor can then manipulate the elements on the board by using his finger in the



mouse, Items can be dragged, clicked and copied and the lecture can handle notes which can be transformed into text and saved. This type of white board is now common in UK schools. An interactive white board can be a cost saver as this technology demonstrates how one Biological can provide learning stimuli for a whole classroom. This is more cost effective than equipping every student with a laptop. Any application that runs on your Biological can be used on an interactive board. It allows to focus the attention of your evidence on a specific area of the screen. The magnifier enlarges a certain area of the screen if necessary. The cost of interactive board ranges from Rs. 10,000/- to Rs. 50,000/- according to its size and quality.

The smart Board is an interactive white board that uses touch detection for user input (Digital visin Touch Technology)

Uses of interactive White board / Smart Board

1. In most of the schools in UK, interactive white boards replaced traditional blackboards, video, DVD players.
2. It is an effective tool for brainstorming.
3. We can add note and drawings and then save them.
4. Show pictures and educational videos to the whole class.

5. You can label parts or highlights elements of an image.
6. This type of tool promotes creative teaching and motivate students to absorb information.

5.4.6. Online Teaching Resources

Many schools and colleges are rapidly changing to **online** learning in response to Covid-19. This topic will **provide** you with the resources teachers need to adopt and **provide** exceptional online instruction for your students. This **guide** will help for teaching and learning can easily adopted for online learning.

Instructional practices for online learning

- During online lessons, maintain a balance between instructional time and students practice. Consider allowing online class time for students to complete work that you would normally assign for home work.
- Encourage students to keep a Physical planner to keep track of assignments and class schedules. During online learning, students are getting a lot of input with instruction and assignments. Writing down key information in their planners will force them to process and prioritize the tasks at hand.
- Well designed Learning content and success criteria are still key in guiding student learning.
- Before you begin to plan a lesson, visualize what you want your students to know, understand and be able to do once that lesson is completed. (Objectives) A pilot of an Airplane cannot plan the flight route until the destination has been determined.
- Determine the students prior knowledge and where the student falls in the learning process.

Assessment in online learning

There are some new factors you may have to consider as you plan assessment for online learning, such as students' interest connection and learning resources (Text book etc.)

- Students will have all of their learning resources at their finger tips. Design assessment that measure higher level thinking and students' ability to transfer their learning.
- Use online tools to collect students' self assessment throughout each unit. Because you are not meeting with students face to face, this feedback will be crucial in your formative assessment process.

Design assessments within the program and asking for students to share all deliverables through the system.

Feedback in online Learning

- Students can review feedback as soon as the teacher posts it on the online platform.
- Individualized feedback can actually be easier and more effective in online tools.
- Schedule mini conferences with individual students throughout each week. Communicate students where they are got in the learning journey. Their self-assessment will help guide you to give them appropriate and needed feedback.

Online Resources for Teacher

Here is a list of free online resources for teachers.

Few Teacher Resources

- Google's Teach from Home
- Google for Education

- DBS resources and lesson plans
- Free access to Brain Pop
- Free access to Newsela
- Discovery Education
- NSTA (National Science Teachers Association)
- Educreations
- Smithsonian's History Explorer
- Khan Academy

Connecting with other Teachers on online

Finally it is important to remember that "You Are Not Alone" during this time! You can find new resources and support from fellow teachers online with the following resources.

Twitter / Social Media Hashtags

- # ed chat
- # cov19edu
- # teaching during corona virus
- # ed leader

Face book Groups

- Amazing Educator Resources
- Educator Temporary School closure for online Learning

Uses of Online Resources

1. The internet is a great resources for learning.
2. Resources are helpful to aid learning and improve teaching.
3. Resources leads to an improved student experience.

4. Online resources help to prepare lesson plan for teachers.
5. Helps to download additional support materials for teaching such as Photos, Videos, Documents, Maps etc.
6. Enhance lectures with multimedia web resources.

5.5. Community resources:

A community resource is anything that has potential to improve the quality of student. We can run innovative projects and activities from these resources. It helps to create a team where we value, support and serve one another.

5.5.1. Zoological Garden

Zoological garden is a place in where animals are housed within enclosures, displayed to the public, and in which they may also breed.

London Zoological Garden which was opened for scientific study in 1828 and to the public in 1857. The number of major animal collections open to the public around the world now exceeds to 1,000, around 80 percent of them are in cities with natural environment. These gardens exist as part of the forest. In these gardens animals like tiger, lion, bear, camel, elephant and varieties of birds lives in a natural environment. Endangered species are conserved in the gardens.

The main importance of zoos is their ability to educate visitors and impart a connection to wild animals, that they display wild animals primarily for the conservation of endangered species, as well as for research purposes and education, Zoos support a number of in situ conservation projects and contribute to valuable research and practical

conservation efforts worldwide, and secondarily for the entertainment of visitors. Some zoos keep animals in larger, outdoor enclosures, confining them with fences, rather than in cages, allow visitors to drive through them and come in close proximity to the animals. Sometimes, visitors are able to feed animals through the car windows. Visitors can see both indigenous and foreign animals in a same place. Animals and birds are kept in the cages with a well maintained climatic condition.

Zoological garden in Tamilnadu

1. Arignar Anna Zoological Park, Vandalur, Chennai.
2. Snake park ,Chennai.
3. Crocodile park, Chennai.
4. Amirthi Zoological Park,Vellore

5.5.2. Botanical Garden

A botanical garden or botanic garden is a garden dedicated to the collection, cultivation and display of a wide range of plants labelled with their botanical names. It may contain special list of plant collections such as cacti and other succulent plants, herb gardens, plants from particular parts of the world. The botanic garden may be an independent institution, a governmental operation, or affiliated to a college or university. If a department of an educational institution, it may be related to a teaching program.

"Botanic gardens are institutions holding documented collections of living plants for the purposes of scientific research, conservation, display and education."

Botanical gardens exchanged plants through the publication of seed lists . This was a means of transferring both plants and information between botanical gardens. This system continues today, although the possibility of genetic

piracy and the transmission of invasive species has received greater attention in recent times

In these Gardens, greenhouses, shade houses, can be found with special collections such as tropical plants, alpine plants, herbs, endangered plant varieties or other exotic plants. These gardens are maintained by experienced staffs.

More than 1800 botanical gardens were situated in the world. The botanical gardens of the 16th and 17th centuries were medicinal gardens. Later, in the 18th century, they became more educational in function, demonstrating the latest plant classification systems. Then, in the 19th and 20th centuries, the trend was towards a combination of specialist and eclectic collections demonstrating many aspects of both horticulture and botany. These gardens attract about million visitors a year, so it is hardly surprising that many people gained their first exciting introduction to the wonders of the plant world in a botanical garden. Botanical gardens are those beautiful places in which you can see various plants from variety of species. These plants are mainly growing for the education and research purpose. Indian botanical gardens are rich with trees. In India most attractive botanical gardens are situating in different areas. Indian Botanic Garden, Howrah in Kolkata is considered as the largest botanical garden in India.

Top 5 Famous Botanical Gardens in India

1. Acharya Jagadish Chandra Bose/Indian botanical garden, Kolkata
2. Garden of the Agri-Horticultural Society of India, Kolkata.
3. Lal Bagh, Bangalore

4. Government Botanical Garden, Ooty
5. Brindhavan Garden, Mysore.

Some important Botanical Gardens in Tamil Nadu

1. Government Botanical Garden, Ooty
2. Semmozhi Poonga Chennai
3. Tamil Nadu Agriculture university botanical garden, Coimbatore.
4. Sims Park, Coonoor, Ooty
5. The Bryant Park, Kodaikannal.

5.5.3 Eco Park

Eco Park is an area located in the interior of a population to create awareness among the rural and urban people about the environment, to live an eco friendly life with lawns, gardens and trees serving as a place of leisure and recreation for the citizens. Referred to as parks also a private or protected, in different ways, where recreational activities are held. The parks are public green areas. Its areas generally having abundance of trees and plants, grass and various facilities (such as benches, playgrounds, fountains and other equipment) that allow to enjoy the Leisure and rest.

Objective of having Ecopark:

1. To conserve the natural environment and to preserve the cultural heritage of that place.
2. To make the public to visit the ecoparks and to understand and enjoy the nature.

People can be entertained by the greenery of plants, meadows, shrubs, medicinal plants, flowers, birds, beetles, butterflies in the park around the year. Ecological parks help to develop research and scientific studies. An ecological park, therefore, helps to generate valuable knowledge about animals, plants and the rest of the members of the ecosystem in question.

Eco Parks in Tamil Nadu

1. Eco Park. Chetpet, Chennai.
2. Tholkappiar Park/ Adayar Park, Chennai.
3. Coutrallam Eco Park, Coutrallam.

5.5.4 Aquarium

Aquarium is a transparent tank of water in which live fish and other water creatures and plants are kept.

An aquarium not only beautifies the living spaces, but also has a therapeutic use that helps to deal with everyday life better. The size of the aquarium does not really matter. Just a small aquarium with some pebbles, plants and fish will serve the purpose. The best place to keep an aquarium is a place where you can look at it for a while every day. Some benefits of aquarium are Reduces stress, helps Alzheimer's patients, Eases pain, decreases anxiety.

Fish tanks are available along with the accessories. Fish food is also available in the market. An aquarium must be maintained regularly to ensure that the fish are kept healthy. Daily maintenance consists of checking the fish for signs of stress and disease. Also, make sure that the water has a good quality and it is not cloudy or foamy and

the temperature of the water is appropriate for the particular species of fish that live in the aquarium.

In 1853, the first aquarium was launched in England as a public aquarium in the London Zoo which came to be known as the Fish House. Government and private organizations are running fish aquarium for the public.

Important aquariums in India

1. Bangalore Aquarium, Bangalore.
2. Calcutta Aquarium, Calcutta
3. Dr.A.M.Michel Aquarium, Kochi.
4. Marine Biological Research Station, Ratnagiri, Maharashtra
5. Marine life Aquarium, Chennai.
6. Aquarium at Chetpet Eco Park, Chennai

5.5.5. Science exhibition / Fair

Many times it is not possible to go to the places for experiencing the reality and gaining first-hand experience. As an alternative the teacher may help the students by gaining useful experiences through the observation and organisation of educationally significant exhibition. One may have a museum or an exhibit soon established in some corner of the institution or may occasionally plan for the exhibition of the things, events and process related with the general education and curricular issues of the school subjects.

The main objectives of organizing science exhibitions are promoting interest in Science and Technology among students and encouraging scientific and technological

creativity among students and inculcating a sense of pride in their talent.

An exhibit is something which one sees as a spectator or visitor at exhibitions. When it is not feasible to go into the community, then the outside world is brought into the classroom in the form of exhibits and concrete representation of things. Exhibits consists of working models, charts, replicas, specimens etc. in which one is not involved in handling it or making it work but remains a passive observer.

Exhibits are seen by parents and public and they congratulate and encourage the skill of the students. Hence there is a chance for these students to do higher education.

Objectives of Exhibitions

1. Science exhibitions bring the school quite close to community and the society.
2. Science fairs help in providing satisfactory forum and opportunity for the healthy competition on the individual and institutional level.
3. These fairs provide opportunity to the talented students for the nourishing and nurturing of their creative talents.
4. Science fairs interest in the students for science.
5. Science fairs gives an opportunity to the students and visitors to understand the practical application and utility of science.
6. To encourage students to make new models, instruments, experiments related to what they had studied in the class room.
7. To identify the junior scientists and congratulate them

Method of organising Science exhibition

Generally science exhibition / fairs are conducted every year in Schools, Region, District, State and National Level. The following procedures are to be followed by the organising committee.

1. Planning
2. Distributing the work
3. Implementation
4. Analysis of the exhibits

Let us see them in brief.

1. Planning

The following points should be kept in mind while planning.

- Objectives of Science exhibition
- Limitations -
 - ✓ Whether our own school exhibits are enough?
 - ✓ Can we include exhibits of neighbouring schools?
 - ✓ What types of exhibits are to be included?
- Working methods to be followed.
- Place, time and date of the exhibition
- Inaugural day and valedictory day celebrations - method of organising.

- Selecting members to Analysis exhibits, no. of prizes required and nature of prizes
- Fund required to organise the exhibition and how to get it.
- Other facilities required to conduct the exhibition.

2. Distributing the work

The following committee has to be framed to conduct the exhibition.

- Committee to give suggestions
- Administration committee
- Committee for entering the events
- Welcoming Committee
- Committee to exhibit things in order
- Advertising Committee
- Helping Committees

According to nature of work, number of committee members may be appointed by the exhibition committee. Students should also be nominated in the above committees according to their knowledge and skill.

3. Implementation

Members of the exhibition committee encourage and motivate all committee members to do their work in time and in a best manner.

Before inauguration of exhibition, all exhibits must arrange in their place and should give self explanation. To explain about exhibits trained students should be arranged on selected items students from surrounding schools and public are to be called to visit the exhibition. Important personals are to be invited for inaugural and valedictory functions.

4. Analysing the exhibits

To analyse the exhibits the following points is to be considered.

Characteristics	Marks
Science Approach	30%
Self nature	20%
Technological work	20%
Completeness	10%
Clear explanation, Nature of exhibits	10%
Interview with the exhibitor	10%

Committee has to analysis each exhibit and award marks and select best exhibits. After the committee's visit students and public can be allowed to see the exhibition.

Next day of the science exhibition, all committee members should sit together and discuss the pros and cons of the exhibition and enter its feedback on the science exhibition Register for future reference.

5.5.6. Fieldtrips

A field trip is nothing but an on-the-spot study of the subject by bringing the students face-to-face with reality. It is a bridge between the school and the community. There are numerous opportunities for the school to utilize the

community resources. By field trips students get first-hand experience with phenomena and aspects which cannot be brought into the classroom.

Learning experiences which are real life and subject to scrutiny and verification are provided for in field trips. Field trips are a means of providing direct experiences, breaking down the monotony of being within the four walls of a classroom, for recreation, for developing keen observation, for collecting concrete experiences/evidences, for bridging the gap between theory and practice.

Field trips arranged situations in the form of educational tours, trips, excursion and outings may provide valuable opportunities to the students for giving real life direct experiences for the learning of many concepts and gaining useful information, knowledge and skills related to various issues of school curriculum.

Using one's sense to understand and get real experience about movable or unmovable, natural or artificial, living or non living things related to subject matter is called field trip. Places selected for field trips should be related to the subject of the syllabus.

Steps taken to follow fieldtrip are

- i. Planning
- ii. Preparation
- iii. Execution
- iv. Evaluation

i. Planning

This is the first and prime stage in planning. If planning is Crystal clear, trip will become very useful. The following should be consider in planning.

- Places to be selected for field trip
- No. of students and teachers participating in the field trip
- Permission sought from officials and school authorities.
- Copy of the programme of the field trip, to be given to parents.

ii. Preparation

- It is the duty of the teacher who is organising the field trip, to prepare all activities in order.
- He can also involve other teachers and student leaders who are accompanying the trip/
- Organiser has to consider the capabilities of teachers and students to allot work to them.

iii. Execution

- Mode of transport has to be arranged in time
- While performing the trip and the places to visit, discipline must be maintained and treat students with controlled independence.
- Questions asked by the students should be promptly answered considering their age and experience. Students should be motivated to ask questions.

iv. Evaluation

After completing the field trip evaluation is to be done within two or three days. In evaluation find whether

- Objectives of the field trip is fulfilled?

- What the merits and demerits are of the field trip.
- Students should be asked to submit their thoughts, knowledge and experiences of the field trip.

Places of visit for field trip

Few field trips places are given for your notice

- Honey extracting centres
- Sanctuaries
- Zoos
- Aquariums
- Power Plants
- Science Laboratories
- Cold storage places
- Industries

Advantages of Fieldtrips

1. They provide rich and meaningful experiences to the students.
2. They provide real life like experiences in planning, executing, observing, analyzing, interpreting etc.
3. Knowledge that the students gain through in field trips is a good exercise and can be retained for a longer period.
4. By practical application of the Biological concepts to real life situations the students are made to realise the true value of Biological Science.
5. Fieldtrip link between school and community.

5.6 New Emerging Media

With the emergence of online technology, the production, distribution and consumption process of news has been changed. Newspapers and television channels

have shifted their importance to the online editions. Each and every media houses in India now have their own websites. Especially the electronic communication with the internet driven technology has brought lots of innovations. The new forms of media have created opportunity for interactions between the producers and consumers of news. These new forms of media are known as New emerging media.

5.6.1 Tele-conferencing

Tele-conferencing is essentially a alive interactive audio, video or audio-video meeting that ensures between two or more participants who are at different places. Here participants communicate via telecommunicating networks using their tablets, mobile phones, laptops, computers and specially designed tech-enabled meeting rooms.

There are different types of teleconferencing. They are

1. Audio Teleconference
2. Audio Graphics Teleconference
3. Computer Teleconference
4. Audio - Video Teleconference

Let us see them briefly

1. Audio Teleconference

This is also called conference calling. Participants links vis telephone lines. Meeting can be conducted via audio conference.

Distance learning can be conducted by audio conference.

2. Audio Graphics Teleconference

This telecommunication channels transit visual information such as graphics, documents and video pictures as an help to voice communication.

This type of communication is useful for meetings and distance learning.

3. Computer Teleconference

Here telephone lines are connected with two or more computers. Anything that can be done on a computer can be sent over the lines. Using electronic mail (Email) memos, reports, letters, lessons can be sent to anyone on the local area network (LAN) or wide area Network (WAN). Items generated by computer which are normally in printed form are sent by E-mail.

Computer conferencing is an emerging area for distance education. Students receive texts and work books via E-mail. Students download these files and use for further study.

4. Audio - Video Teleconference

Voice communication and Video images are used for conferences. It can display anything that can be captured by a TV camera. Here you can see and hear person on a television monitor. Participants face-to-face meeting enables to see the facial expressions and physical behaviour of participants at remote sites Graphics are used to enhance understanding.

Video conferencing is an effective way to use one teacher two teaches to a number of sites. Video conferencing maximizes efficiency because it provides a way to meet the several groups in different locations, at the same time.

Tools needed for Video Teleconferencing

To get started with Video Teleconferencing and connect your sites, you need the following few things.

- A desktop computer, laptop or smart phone
- An internet connection
- A microphone and webcam
- An external speaker
- Video Telecommunication software

Advantages of Teleconferencing

1. Saves Time :

The most important advantages of Teleconferencing is time-saving. It is possible to hold the meeting on a very short notice. In addition there is no time spent to travel to the venue of the meeting as all meetings are held through internet and computers.

2. Save on travel expenses :

Now-a-days travel costs can be quite significant. Teleconferencing removes the need to travel and saves huge money.

3. Efficient Record Keeping :

One of the main advantage of Teleconferencing is efficient meeting Record keeping. Computer devices are able to record, keep every details of the meeting. Teleconferencing also makes it very easy to retrieve this data in the future and make references whenever necessary.

4. Cut conference costs :

There are many costs to hold a meeting. Hiring venue, buying food and refreshments etc. associated with any

meeting. Teleconferencing removes all these costs as no such arrangements need to be made.

5. Reliability :

Teleconferencing is one of the most reliable ways of holding meeting. Due to advancement in technology, teleconferencing channels are today much more stable. It is also very secure mode of communication where the safety of data as well as privacy is guaranteed.

6. Services :

Health care management, Business, Education, International conference and on line interviews are held using Teleconferencing.

5.6.2 Communication Satellites

Communication satellite is an artificial satellite that relays and amplifies radio communication signals via transponder; it creates a communication channel between a source transmitter and a receiver at different locations on earth. Communication satellites are used for television, telephone, radio, internet, Metrology and Military applications all over the world. As on 1st January 2021, there are 2224 Communication satellites in Earth orbit. Most communication satellites are in geostationary orbit 35,900 kms above the equator so that the satellite appears stationary at the same point in the sky.

Communication satellites use a wide range of radio and microwave frequencies. To avoid signal interference, international organisations, have regulations for which frequency ranges or 'bands' certain organisations are allowed to use. This allocation of bands minimizes the risk of signal interference. These satellite provide communication links between various points on Earth.

Watching your TV shows or movies from internet would have been impossible without this.

Satellites and Education

India launched Education Satellite GSAT-3 known as EDUSAT on 20th September 2004 from Sriharikota, India. It is meant for distant classroom education from school level to higher education. This was the first dedicated Educational satellite that provide the country with satellite based two way communication to classroom for delivering educational materials.

Objectives of EDUSAT

- Providing support to formal and non-formal education.
- Teachers' training program
- Increasing access to quality resource persons
- Enhancing community participation
- Taking education to remotest corner of the country.

Application of EDUSAT

- Conventional Radio and Television broadcasting
- Interactive Radio and Television
- Exchanging of Data
- Teleconferencing both one way or two way & Audio conferencing.
- Computer conferencing
- Web based education/

EDUSAT Worlds' first satellite meant only for educational purposes is being used. Training by teleconferencing overcomes the problem of transmission loss since

participants are watching, listening and interacting with primary resource persons. CIET a Constituent unit of NCERT using this technology of teleconferencing over a decades to give training / orientation to teachers and teacher educators. Other institutions like IGNOU, Vigyan Prasar, Sarva Shiksha Abhiyan (SSA), Rehabilitation Council of India (RCI) etc are using EDUSAT.

Advantages of Communication Satellites

1. Flexibility
2. Ease in putting in new circuits
3. Distances are effortlessly taken care of and expense does not make a difference.
4. Broadcasting Conceivable Outcomes
5. Each and each side of the earth is secured.
6. User can control the system.

Disadvantages of Communication Satellites

1. The introductory costs are excessively high
2. Congestion of frequencies
3. Interference and proliferation

5.6.3 Computer Networking

A computer network is a set of computers sharing resources located on or provided by network nodes. The computers use common communication protocols over digital interconnections to communicate with each other. These interconnections are made up of telecommunication network technologies, based on physically wires, optical, and wireless radio-frequency methods that may be arranged in a variety of network topologies.

The nodes of a computer network may include personal computers, servers, networking addresses, and may have host names. Host names serve as memorable labels for the nodes, rarely changed after initial assignment. Network addresses serve for locating and identifying the nodes by communication protocols such as the internet protocol.

Computer networks may be classified by many criteria, including the transmission medium used to carry signals, bandwidth, communications, protocols to organize network traffic, the network size, the topology, traffic control mechanism and organizational intent.

Computer networks support many applications and services such as access to the world wide web, digital video, digital audio, shared use of application and storage servers, printers and fax machines and use of email and instant messaging applications.

A computer networks extends inter personal communications by electronic means with various technologies such as email, instant messaging, online chat, voice and video telephone calls and video conferencing. A network allows sharing of network and computing resources. Users may access and use resources provided by devices on the network such as printing a document on a shared network printer or use of a shared storage device. A network allows sharing of files, data and other types of information giving authorized users the ability to access information stores on other computers on the network. Distributed computing uses computer resources across a network to accomplish tasks.

Types of Computer Network

A computer network is mainly of four types. They are

1. LAN (Local Area Network)
2. PAN (Personal Area Network)
3. MAN (Metropolitan Area Network) and
4. WAN (Wide Area Network)

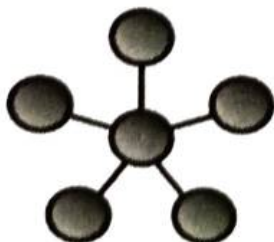
Need of computer networking

Computer networks help users on the network to share the resources and in communication. Networking of computers help the network users to share data files. Users can share devices such as printers, scanners, CD-Rom drives, hard drives etc.

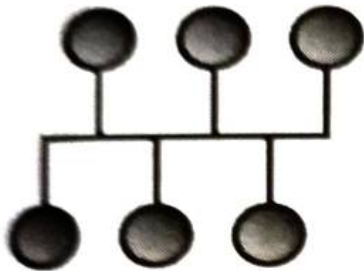
Types of network Topology

Topology is derived from two greek words topo and logy where topo means place and logy means study. In computer network topology is used to explain how a network is physically connected. A Topology mainly describes how devices are connected with internet and other communication links. Network topology defines the layout, virtual shape or structure of the network. Types of network topology are

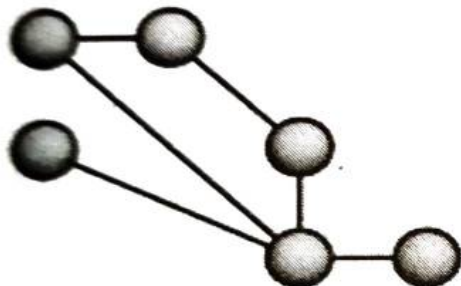
1. Star Network



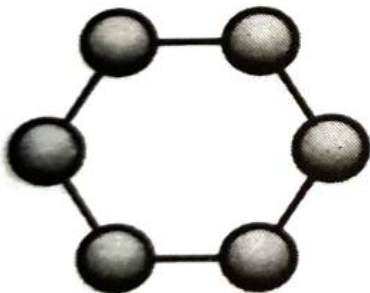
2. Bus Network



3. Mesh Networking



4. Ring Network



Advantages of Networking

1. Strengthen business connections
2. Get fresh ideas
3. Raise your profile
4. Get access to job opportunities
5. Gain more knowledge
6. Get Career advice and support

7. Built confidence
8. It boosts storage capacity of the computer
9. It makes file sharing easier
10. It is an inexpensive system

Disadvantages of Computer Networking

1. The connected systems on a network entirely depend on the main server. If there is any problem in server, complete network failed.
2. Since networks are based on the centralized server, most of the decisions are made themselves.
3. Setting up Computer network can be expensive
4. Due to interactions, a virus can easily spread between computers in a network.
5. In networking, large number of users are using computers. Hackers can easily access the network using specialized tools.
6. It can drastically decrease the productivity of the company.

5.6.4 Word Processors

Word processor is a computer program used to write and revise documents, compose the layout of the text and preview on a computer monitor how the printed copy will appear. The last capability is known as "what you see is what you get".

Word processors facilitate writing and editing, especially with their ability to copy and move text, (Cut and paste) their built-in dictionaries to check spelling, and their grammar checkers. Other common features include a wide choice of typographic fonts and sizes, various paragraph and page

layouts, tools for finding and replacing strings of characters and word counts. Modern word processors also have many features one reserved for desktop publishing systems, such as table creation and importation of graphic images. They typically provide templates for common document types, such as letters, memos and resumes and can generate multiple copies of a document with recipient addresses drawn from a list. (Mail merge)

Desktop publishing programs have word processing features but also provide highly flexible layout and control over appearance to combine text and graphics for advertising copy, magazines and books.

Examples of Word processor softwares

- Abiword
- Apple iwork-pages
- Apple Text Edit
- Coral word perfect
- Dropbox paper
- Google Docs
- Libe office → writer
- Microsoft office → Microsoft word

Functions of word processors

The functions of word processing described below are all basic ones if you have reasonably computer literature.

- Creating, editing, saving and printing documents.
- Pasting, moving and deleting text with in a document.
- Formatting text such as font type, size, bolding, underlining or italicizing.

- Creating and editing tables
- Adding symbols.
- Changing the text appearance
- Changing the page appearance

Advantages of Word processors

1. A document can be stored in a computer for a future reference.
2. Have special editing tools namely spelling and grammar checkers.
3. One can easily insert or replace a word or phrase without affecting the neatness of a document.
4. You can save your work and come back to it at a later time.
5. Plenty of professional quality document templates available online.
6. Can import a data from a database and use it to create mail merge.
7. Teacher and students gain a sense of security about losing assignments.
8. Teacher benefit by receiving a readable copy that is easy to grade for students.
9. Very helpful for publishing work
10. Work done on a word processor and saved on the internet is highly portable and accessible from anywhere.
11. Environment improves as there is no use of paper.
12. It helps to reconstruct complex office work.

disadvantages of Word processors

1. Cost of word processor is very high.
2. Most users will probably never use 50% of the functionality of word.
3. Some functions are not always intuitive and it can take a while to get your desired effects.
4. You always need an internet connection to access your data files.
5. Security measures can be breached.

5.6.5 Blended learning

1. Introduction

Blended Learning is about effectively integrating ICTs into course design to enhance the teaching and learning experiences for student and teachers by enabling them to engage in ways that would not normally be available in their usual environment, whether it is primarily face-to-face. Blended Learning technologies are as follows:

- Broaden the space and opportunities available for learning.
- Support course management activities.
- Support the provision of information and resource to students.
- Engage and motivate students through interactivity and collaboration.

The learning and teaching activities need to be meaningful and relevant for the student learning.

Definition:

- The thoughtful integration of classroom face-to-face learning experiences with online learning experiences.
- It combines online with face-to-face learning. The goal of blended learning is to provide the most efficient and effective instruction experience by combining delivery modalities.
- Mixed mode or hybrid-learning is the integration of face-to-face (F2F) learning with online learning activities.
- A solution that combines several different delivery methods, such as collaboration software, web-based courses, EPSS and knowledge management practices.

Objectives

Increase the amount and quality of faculty-to-student and student-to-student interaction.

- Increase opportunities for active and collaborative learning assessment before, during and after lectures.
- Help students prepare for class discussions or lab work.
- Facilitate more varied and engaging media for presenting course content.
- Address learning bottlenecks via new types of interactive and independent learning activities.
- Drive collaborative teacher problem-solving.

Uses :

- Blended learning can increase access and flexibility for learners, increase level of active learning and achieve better student experiences and outcomes. For teaching

staff, blended learning can improve teaching and class management practices.

- **Enhanced learning:** Students usually receive more feedback and more frequent feedback, for their instructors. Students can acquire useful skills from using the Internet and computer technology (live instructor presence, technological enhancement). Students have more time to reflect and refer to relevant course and other research materials when working and writing online than when responding in class.
- **Support for student collaboration:** Real-time real-space interactivity and asynchronous online interactions, synchronous online interactions.
- **Course accessibility:** Students have access to unlimited up-to-date resources available via the web. Live signing in live courses, transcription of audio and video, labeling of images with alt text online.
- **Learner convenience:** Students have greater time flexibility, freedom and convenience by working part of the time online. Time flexibility without full-term scheduling, a synchronicity (Non-real-time learning), and the repeatability of some automated learning and ease of accessing course materials.
- **Rich learner assessment:** Automated assessments, live instructor-led assessments, peer assessments live or via the online classroom.
- **24/7 Accesses:** Students often develop or enhance skills in time management, critical thinking and problem solving. Students typically have 24/7 access to online course materials.

- **Multi-use/Dual use resources:** Course contents may often be used in both contexts (F2F and online)

5.6.6 Flipped Classroom

Flipped Classroom is an approach that allows teachers to implement a methodology or various methodologies in their classrooms. It means "School work at home and home work at school". Flipped classroom inverts traditional teaching methods, delivering instruction online outside of class and moving "homework" into the classroom.

Definition

Flipped Classroom is a pedagogical approach in which direct instruction moves from the group learning space to the individual learning space and the resulting group space is transformed into a dynamic, interactive learning environment where the educator guides students as they apply concepts and engage creativity in the subject matter.

It is called flipped because the traditional definition was "what used to be class work (lecture) is done outside the class (usually with videos at home) and what used be home work (assigned problems) is now done in the class".

The flipped classroom is a pedagogical approach in which typical lecture and home work element of a course are reversed. Short video lectures of instructional content are viewed by students at home before the class session, while in the class time is developed to solve problems exercised, projects or discussions with the guidance of the teacher.

Frame work of flipped classroom

It is a student centred model in which class time explores topics in greater depth and create meaningful learning opportunities. In a flipped class room content

delivery may take a variety of forms. Video lessons prepared by the teachers or others are used to deliver content although on line collaborative discussions and text readings may be used. More time can be spent in class on higher order thinking skills such as problem - finding, collaboration, design and problem solving. Students take different problems, working in groups construct knowledge with the help of their peers and teachers. Teachers interaction with students in a flipped classroom can be more personalised and less didactic.

Flipped classroom apply a mastery learning model that requires each student to master a topic before moving to the next one.

Four pillars of Flipped classroom

- F – Flexible Environment
- L – Learning Culture
- I – Intentional Content
- P – Professional Educators

1. Flexible Environment

The flipped classroom must allow for a variety of learning models. Educators will often physically earning their learning space to accommodate the lesson unit. Additionally flipped educators have to be comfortable with flexible learning environment in which the student choose what and where they learn outside of the classroom.

2. Learning Culture

In the flip learning model, educationists are more cognizant of the content and they choose to deliver and

determine what direct instruction or traditional lecture can be shifted. Students are empowered to find content on their own, while educator help them make connection and buildings on their own understanding are proficiency. Parents and community must be kept engaged and informed for the culture to shift in this new model.

3. Intentional Content

Educators must continuously think about how they can use the flip learning models to help the student's gain conceptual understanding as well as procedural fluency when needed. They must evaluate what they must need to teach and what the materials students should explore their own. Educators must adopt various methods of instruction such as active learning strategies, peer instructions, problem based learning or mastery or Socratic Method depending on grade level and "Subject Matter".

4. Professional Educators

The role of the teacher is even more important and it is often more demanding in flipped classroom than in a traditional one. During class time, teacher continually observe their students, providing them with feedback relevant in the movement and assessing their work. There is a deliberate shift from lecture centered to a student-centered class. Also educators must be able to connect with each other to improve their trade, to accept constructor criticism and to tolerate controlled classroom chaos.

Flipped classroom

The flipped classroom describes a reversal of the traditional teaching where students gained first exposure to new material outside of class, usually via reading or lecture, videos and then class time is used to do the harder work of

assimilating that knowledge through certain strategies such as

1. Problem – solving
2. Discussion and
3. Debates

Advantages

1. Students take control of their learning.
2. Student can develop foundation of factual knowledge.
3. Students can understand facts and ideas in the context of a conceptual framework.
4. They can enrich their knowledge in ways that facilitate retrieval and application.
5. It saves time. Teacher may spend more time with students.
6. Students learn at varying speed.
7. Ability to rewind, save, review lesson plans at any moment.
8. Students are provided opportunities for review lesson front-load for classroom activities.
9. Patents can view lessons and better assists students.
10. Students do not struggle with completing homework because they forgot how it does.
11. Students are actively working with their peers.
12. Students take ownership of their learning.

13. Personalized educations for slow and fast learners.
14. Stronger bonds with teachers.
15. It provides adequate learning opportunities for verbal learners and for visual learners.

5.6.7 Artificial Intelligence

Artificial Intelligence (AI) broadly refer to any human-like behaviour displayed by a machine or system. In AI most basic form, computer are programmed to 'mimic' human behaviour using extensive data from past example of similar behaviour. With AI, machine can work efficiently and analyse vast amounts of data in the blink of an eye, solving problems through supervised, unsupervised or reinforced learning.

AI is now part of our daily lives. We have AI solutions for quality control, video analytic, speech-to-text and autonomous driving, as well as solutions in healthcare, manufacturing, financial services, entertainment and education.

History of Artificial Intelligence

Before 1949, computers could execute commands, but they could not remember what they did as they were not able to store these commands. In 1950, Alan Turing discussed how to build intelligent machines. Research done in this field from 1960 and laboratories had been established around the world. By 2000 solutions developed by AI researchers, were being widely used.

Application of AI

Artificial intelligence, the intelligence exhibited by machines has been used to develop thousands of applications to solve specific problems throughout industry and academic.

Here are some of the real-world applications of AI in Education.

1. Automation of basic administrative activities

A lot of time is spent by teachers in administrative activities like grading and assessment of worksheets. The use of AI in education can help automate the grading and assessment of activities like multiple choice questions, fill in the blanks etc. Another tedious and cumbersome activity for teachers is preparing the report cards of students. The use of AI in education can help automate this as well. Automation of administrative activities means teachers can spend more time with the students, thus making the learning process more efficient.

2. Personalized learning

The purpose behind application of AI in education is not to replace teachers, but give them a helping hand in understanding the potential and limitations of each student. The use of AI in schools make things easy and convenient for the teachers and students as well. By understanding the needs of every student, teachers can come up with a study plan for every students.

3. Constructive feedback

Programs powered by AI can provide valuable feedback both to the students as well as teachers. Use of AI in Classrooms can point out to teachers how to improve the instructions provided to the learners and also how to make learning more fun and interesting. Instant feedback to students helps them understand where they are going wrong and how they can do it better.

4. Personalization

What better way to offer more personalized learning opportunities for students than to have. AI be able to analyze student responses, determine areas of need and interest, and find resources or create new questions to help students to greater understanding of the content.

5. Communication

Students and teachers will be able to communicate instantly with one another as well as to connect with other forms of AI around the world.

6. Accessible by all students

Due to socio economic status, transportation and disabilities may change how much access a student has to the classroom. However learning can happen anywhere with the help of AI.

7. Educational softwares are developed to student needs

AI gives importance to individualized learning. AI responds to the needs of the student, put greater emphasis on certain topics, repeating things that students have not

mastered and generally helping students to work at their own pace.

8. **AI may change the whole education system**

AI has the potential to change about everything on education. Students can learn from anywhere in the world to anytime and sometimes AI may replace teachers in some instances. Educational programs powered by AI are helping to learn basic skills and they will likely to offer students a much wider range of services.

Advantages of Artificial Intelligence in Education

1. AI-based education can provide an opportunity to study at any time.
2. AI-based solutions can adopt according to the level of knowledge, interesting topics etc. The system tends to help students with their weak side.
3. AI based education offer virtual mentors to track the students' progress.
4. AI-enabled education offers appropriate teachers, depending on the teaching experience and soft skills.
5. AI operates 24 x 7 without interruption or breaks
6. AI augments the capabilities of differently abled individuals.
7. It introduces a new technique to solve new problems.
8. It handles the information, better than humans.
9. It improves work efficiency.
10. It uses more powerful and more useful computers.

Disadvantages of Artificial Intelligence in Education

1. The implementation cost of AI is very high.
2. AI based teaching lead to serve unemployment.
3. Development of software for AI is slow and expensive.
4. It lacks creativity and decreases the thinking power of the students.
5. Human interaction decreases.

5.6.8 Augmented Reality

Augmented Reality is an interactive experience of a real world environment where the objects that reside in the real world are enhanced by computer generated perceptual information, sometimes across multiple sensory modalities, including visual, auditory or other sensory information.

Augmented Reality is a technology in which a computer-generated image is super imposed on to the user's vision of the real world, giving the user additional information generated from the computer model. Using an AR system, the user's view of the real world is enhanced.

The basic idea of augmented reality is to super impose graphics, audio and other sensory enhancements over a real world environment in real time. Some of the most exciting augmented reality work began taking place in research labs at universities around the world. In February 2009, Mistry presented augmented reality system at MIT Media Lab and they called it sixth sense. Basic components that are found in many augmented reality systems are

- Camera
- Small projector
- Smart Phone
- Mirror

The device worked by using the Camera and Mirror to examine the surrounding world, steering that image to the phone (which processed the image, gathered GPS coordinates and pulled data from the internet) and then projected information from the projector onto the surface in front of the user, whether a wrist, a wall or even a person. Because the user was wearing the camera on his chest, sixth sense augmented whatever he looked. For example, if he picked up a can of soft drink in a grocery shop, sixth sense found and projected on to the soft drink information about its ingredients, price, nutritional value - even customer reviews. If the customer wanted to know more about that soft drink, he could use his fingers to interact with the projected image and learn about, say, competing brands.

Uses of Augmented Reality

In business field, Augment will project any new product into a real world environment. Potential buyers can 'see' the product in their own homes and read all information and reviews before making a purchase. AR will also create a fully-immersive 3D map that guides you wherever you want to go.

Health care personals rely on smartphone - enabled AR. A special app that helps doctors and nurses use their phones to quickly identify specific types of wounds for faster diagnosis and more efficient care.

Militaries were some of the early adopters of gaming technology, seeing the possibilities for training soldiers for warfare realistically but in safe settings. And militaries are likely to do the same with AR.

Advantages of Augmented Reality

1. It reduces line between real world and virtual world.
2. It enhances perceptions and interactions with the real world.

3. Due to its use in medical industry, life of patients have become safer. It helps in efficient diagnosis of diseases and in early detection of them.
4. It can save money by testing critical situations in order to confirm their success without actually implementing in real time. Once it is proven, it can be implemented in real world.
5. It can be used by military people without putting their life in danger by way of battle fields simulation before the actual war. This will also help them in actual war to take critical decision.

Disadvantages of Augmented Reality

1. It is expensive to develop the AR technology based projects and to maintain it. More over production of AR based devices is costly.
2. Lack of privacy is a concern in AR based application.
3. In AR, people are missing out an important moments.
4. Low performance level is a concern which needs to be addressed during testing process.
5. It requires basic learning to effectively use AR compliant devices.

5.7 Qualities of a Good Biological Text Book

Text books take an important part in teaching learning activities. Text books help students to attain their goal. A good Biological Science text book not only satisfies the syllabus but also it must be understandable to students. Generally a Biological Science text book should have the following three characteristics.

1. It should have quality in materials used
2. It should be useful to teachers.

3. It should be useful to students

A Good Biological Science text book should have the following properties.

1. Biological Science text book should have long term and short term objectives. From these objectives students should get knowledge, attitudes and skill.
2. Text books should be student centred. There should not be any hindrance in understanding the subject matter.
3. Text book wrappers, quality of paper and size of the letters should create interest in reading the book.
4. Structure of the book should be strong.
5. Content should be written in logical and psychological way. It should be according to Biological Science syllabus.
6. Figures and explanations must be placed in appropriate places. Printing Figures statements and formulae in colour in preferable.
7. Text book should be without any spelling mistakes.
8. It should train three category of students namely backward, average and high achievers.
9. Necessary information must be given to solve the problems on their own.
10. At the end of each unit formative and summative questions of all types should be given.
11. Text books should be written with experienced and qualified teachers.
12. Text books should contain the problems which are used in daily life situations.

Method to select a good book

To select a good book George Hunder recommends the following ways to evaluate the book.



Sl. No.	Characteristics	Max. Marks
1.	Qualities of the Author	50
2.	Structure and cost of book	100
3.	Content formation is psychological way	300
4.	Content	250
5.	Style of the language	110
6.	Learning experiences	140
7.	Uses of text books to teachers	50
	Total	1000

Teachers can buy a book which gets maximum marks.

Uses of Text books to teachers

1. Text books are written on par with syllabus. This helps the teacher as a guide to teach the content.
2. To teach a lesson with plan, to select a problem, to select the teaching method, to select the instructional resources while teaching text book help the teachers.
3. Since text books are written in psychological and logical methods, it help teachers to teach the subject matter in order and in sequence.
4. In a good Biological Science text book all types of theory and problems are written at the end of each unit. If teacher solves these problems and take classes, he will get self confidence in teaching.
5. To give class work, home work and assignments as per the students knowledge, text books are helpful.
6. Problems and few solutions are given in the text books. This help the teacher to reduce the time while preparing similar problems.
7. To prepare Unit test and Achievement tests text books are very useful to teachers.

8. Books are written by experienced teachers. Their experience and knowledge help unexperienced teachers to acquire proper knowledge.
9. Text books are very useful in writing lesson plan.
10. To create lively classroom activities, text books are useful.

Uses of text books to students

1. To recall and read the content done in the class, text books are helpful.
2. Text books are very helpful to revise and practice.
3. It helps for self learning and to solve problems.
4. To read the subject matter well in advance before it is done in classroom, text books are helpful.
5. It helps for meaningful learning.
6. When a student takes leave to school for some reasons text books are used to read the lessons at home.
7. To know the objectives of teaching of Biological Science, to clear the doubts of class teaching text books are used.
8. For deep study of the subject matter text books are used.
9. New information, Tabular Columns and explanations are given in text books. This creates interest in education.
10. To read and understand the text in according to their learning skill of students, text books are helpful.

5.8. Qualities of a good Biological Teacher

"A teacher can never truly teach unless he is still learning himself. A lamp can never light another lamp unless it continuous to burn its own flame".

- Rabindra Nath Tagore

Teaching is one of the essential jobs of a teacher, besides his other involvements with students, inside and outside the classroom, laboratory, library, play ground etc. He has to be a qualification teacher. "A teacher has to help the bud into full bloom and not to make paper flowers to satisfy his whim".

- Sarvapalli Dr. RadhaKrishnan

"The quality of a nation depends upon the qualities of its citizen. The quality of its citizen depends upon the quality of their education. The quality of their education depends upon the quality of their teacher. So the quality of a nation depends upon the quality of its teachers".

- American Commission on Teacher Education

The following qualities are required in a Biological Science teacher besides his academic and professional requirements.

1. Mastery over the subject

A Biological Science teacher should have gained mastery over the subject. For this he has to update his knowledge according to the present situation. He should have the habit of reading Journals and modern books. He has to attain new content knowledge and new teaching methods.

2. Knowledge of Psychology

The teacher should have knowledge of educational Psychology, so that he can handle the students with care. Teacher has to bear in mind that a student learns only that activity which brings his pleasure, success or some reward and conversely student does not learn an activity which brings his failure or punishment and causes him some sort of pain. Teacher should be affectionate and patient towards the pupil.

The teacher should be a guide, helper and friend. The teacher must study the students and should know the laws of learning for which a study of psychology is essential. He should adjust his teachings accordingly.

3. Well Equipped with methodology of Teaching Biological Science

He must have the knowledge of various teaching methods namely Lecture method, Laboratory method, Demonstration method etc. and use it according to the units.

4. Faith in the subject

Teacher's faith helps him to realize the utility and the values of the subject by correlating with every day life. He should love his subject and his students.

5. Well equipped in Human Qualities

He should have a satisfactory mental health and adequate adjustment. He should have qualities like patience, love, respect for his profession, good character, leadership qualities, self confidence and self criticism.

6. Classroom Activities

Teaching should be carried out with well planning. Teaching should link the previous knowledge of students to the new knowledge. He should create research attitude and critical thinking while teaching. He should use Biologicals and blackboard extensively. He should use the teaching aids available in the school. He should write the lesson plan on par with objectives and teach the same to students.

7. Knowledge on various Fields of Biological Science

The teacher should have a fair knowledge of biological science important topics and master in applications of computer.

8. Fluency in expression

Teacher should be impressive and his speech should be very effective and understandable while teaching. He should have the skill of communication.

9. Adequate General Knowledge

A Biological Science teacher has to be all rounder. He should know all current events of everyday life and living. He should have basic knowledge of other school subjects.

10. Well equipped in Teacher Qualities

A Good Biological Science Teacher have coordinational relation with students, fellow teachers, parents and officials. His teaching methods, ways of handling students, classroom activities will tell that he is a good teacher.

11. Punctuality

A Good Biological Science Teacher will go to a class in time. He will submit the achievement test answer papers in time and give progress cards to students in time.

12. Professional Enrichment

For Professional Enrichment, Biological Science teacher should participate in faculty meetings, Science Club, Education Fairs, Biological Science projects etc.

Review Question

1. How print resources are helpful in teaching Biological Science?
2. Explain any two Audio resources to teach Biological Science effectively.
3. How charts and flash cards are helpful in class teaching.
4. Mention ICT Resources to teach Biological Science.